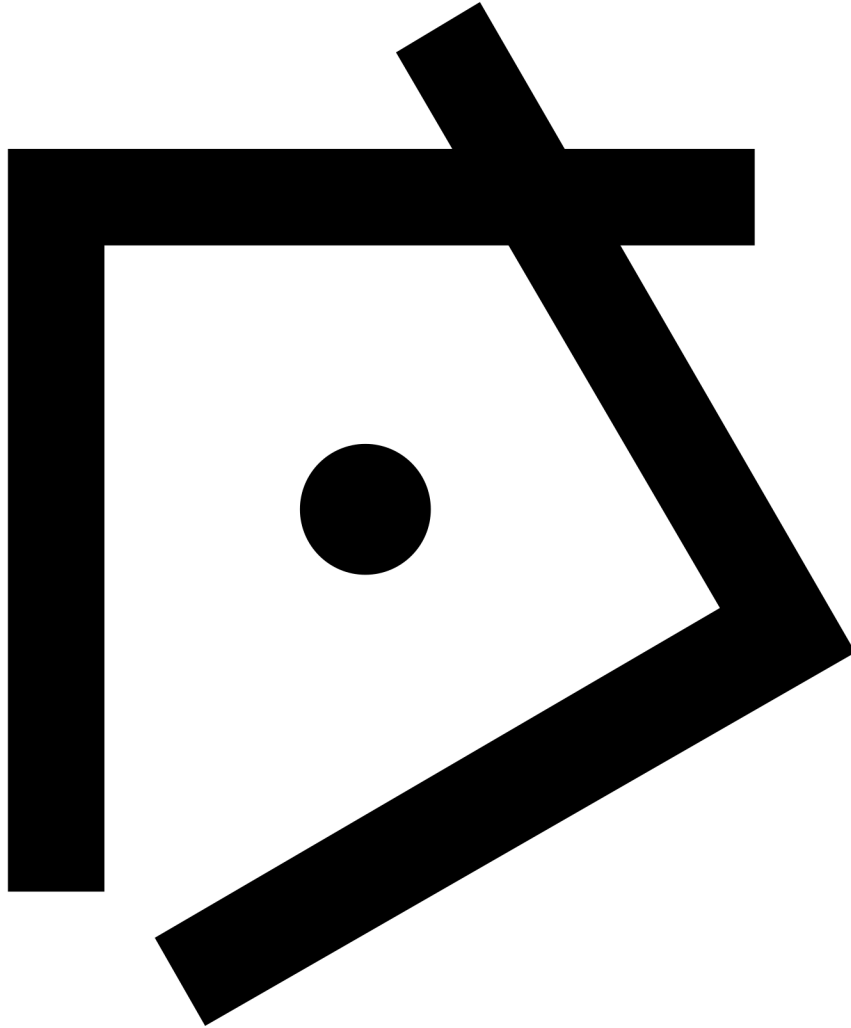


# **QNyx V3.0.2 Operator's Manual**



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August 7 2026

# QNYx Transport - Operator's Manual

## Applies to:

- QNYx v3.0.2 (August 2026)
- Linux

**Audience:** Network technicians comfortable with Linux CLI.

## 1. About This Manual

This manual provides the essential operational procedures, verification steps, and troubleshooting workflows required to manage **QNYx Transport** nodes.

### 1.1 Scope and Exclusions

This document is strictly an operational guide. It covers the day-to-day deployment, configuration, and maintenance of QNYx software endpoints.

- **Included:** CLI command syntax, interface verification, firewall requirements, and log-based troubleshooting.
- **Excluded:** This manual does not serve as a cryptographic reference or protocol specification. Detailed inner workings of the underlying primitives.

## 2. System Overview

- QNYx creates encrypted point-to-point tunnels between a **server** and one or more **clients** over UDP, with a separate TCP channel used only for key exchange (KEX).
- Each connected client gets a TUN interface (qn $\times$ 0, qn $\times$ 1, ...) and a private /30: server side 10.<server\_id>.<net\_id>.1, client side 10.<server\_id>.<net\_id>.2.
- Identity is a keypair (ML-DSA-65). A server trusts exactly the public keys placed in its approved-keys directory - there are no passwords or accounts.
- Session keys rotate automatically every 15 minutes; no operator action is ever required for key management after install.

## 3. Installation

### 3.1 Requirements

- Linux with /dev/net/tun available
- Root (QNYx configures interfaces and, by default, binds TCP 443).

### 3.2 Generate an identity

```
qnyx -g <output_directory>
```

Writes `qnyx.priv` and `qnyx.pub` to the directory. Rename per site convention (e.g. `SiteA.priv` / `SiteA.pub`).

- The `.priv` file never leaves the machine it was generated on.
- The `.pub` file is what you exchange with the other side.

### 3.3 Enroll identities

- On the server: copy each approved client's `.pub` file into the server's approved-keys directory (the `-pu` directory). The directory is re-read on every incoming handshake — adding or removing a client's key takes effect immediately, no restart needed.
- On each client: copy the server's `.pub` file to the client (passed as `-pu`).
- To revoke a client: delete its `.pub` from the server's approved-keys directory. Its current session survives until it next disconnects or rekeys.

## 4. Command Reference

**Server:** qnyx -s -id <0-255> -pr <server.priv> -pu <approved\_pubkeys\_dir> [-kp <port>]

**Client:** qnyx -c <server\_ip> -pr <client.priv> -pu <server.pub> [-kp <port>] [-dp <port>] [-mtu 600-1410] [-r]

**Keygen:** qnyx -g <output\_directory>

| Flag          | Meaning                                                                        | Default                   | Notes                                                                                                      |
|---------------|--------------------------------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------|
| -s            | Server mode                                                                    | —                         | Requires -id                                                                                               |
| -c <ip>       | Client mode, connect to server IP                                              | —                         |                                                                                                            |
| -id <0-255>   | Server identity ID                                                             | —                         | Becomes the second octet of all tunnel IPs (10.<id>.x.x). Must be unique per server across the deployment. |
| -pr <path>    | Private identity key                                                           | —                         | Server's lease file is derived from this path (<path>.leases)                                              |
| -pu <path>    | Server: approved-pubkeys <b>directory</b><br>Client: server pubkey <b>file</b> | —                         |                                                                                                            |
| -kp <port>    | KEX port (TCP)                                                                 | 443                       | Same value must be used on both sides                                                                      |
| -dp <port>    | Data port override (client only, UDP)                                          | random<br>1024 -<br>65535 | Set this explicitly at any site with strict egress filtering, so the firewall rule can be pinned           |
| -mtu 600-1410 | MTU override (client only)                                                     | 1410                      | Set this to accommodate lower MTU links than 1500 bytes                                                    |
| -r            | Route flag                                                                     | FALSE                     | Set this flag on client mode to default route traffic through the tunnel.                                  |

- Log file: /var/log/qnyx.log (epoch-seconds timestamps). If the log file can't be opened, entries fall back to stdout prefixed [LOG\_ERR].

## 5. Network Requirements

What the site firewall must allow:

| Direction       | Proto/Port               | Purpose                                         |
|-----------------|--------------------------|-------------------------------------------------|
| Client → Server | TCP <kp> (default 443)   | Key exchange, initial + every rekey             |
| Client ⇌ Server | UDP <dp> (random or -dp) | Tunnel data - <b>same port number both ends</b> |

- **NAT:** clients behind NAT are supported; the server is not (it must be reachable on both ports). Client-side NAT must permit outbound UDP and its return traffic.
- **MTU:** tunnel interfaces are defaulted at 1410. Max on-wire encoded packet is 1467 bytes at this MTU setting.
- **Recommendation for fixed sites:** restrict inbound TCP <kp> on the server to known client IPs. The KEX port accepts connections from anywhere before authentication; internet scanners hitting it are harmless but generate log noise, and each connection briefly occupies the handshake listener.

## 6. Log Message Reference

Every message QNyx can emit, what it means, and what (if anything) to do. Severity: INFO = normal · WATCH = normal in isolation, a pattern matters · ACT = operator action needed.

| Message                                                                                                      | Severity | Meaning / Action                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [+] TUN interface qnxN created (fd: X)                                                                       | INFO     | Interface allocated. On reconnects the same fd number reappearing is normal and correct.                                                                     |
| [+] Secure tunnel established: P -> IP:P                                                                     | WATCH    | Key exchange succeeded and the data channel is set up. Does not yet prove traffic flows.                                                                     |
| [+] Identity <path> assigned sticky net_id N (10.x.y.1)                                                      | INFO     | Server accepted the client and assigned its address.                                                                                                         |
| [*] Server polling KEX on port P with Server ID N...                                                         | INFO     | Server startup.                                                                                                                                              |
| [*] Loaded identity lease store: <path>                                                                      | INFO     | Server startup.                                                                                                                                              |
| [+] In-band key rotation KEX authorized for net_id N                                                         | INFO     | Scheduled 15-min rekey completed for that client (server side).                                                                                              |
| [+] RX rekey trigger verified; requesting KEX/[*] Rekey trigger received.../[+] Session key securely rotated | INFO     | Scheduled rekey (client side).                                                                                                                               |
| [+] TX switched to new session key epoch                                                                     | INFO     | Node started transmitting under the new key. May appear slightly offset from the lines above; the asymmetry is normal.                                       |
| [-] Client rekey exchange failed: <err>                                                                      | WATCH    | One missed rekey; retried automatically next cycle.                                                                                                          |
| [-] Socket descriptor KEX loop fault: <err>                                                                  | WATCH    | A KEX/rekey TCP connection failed server-side, or a scanner poked the KEX port.                                                                              |
| [!] Session for net_id N lost heartbeat; reclaiming tunnel resources/[*] Released net_id N                   | WATCH    | Server dropped a silent client (no traffic for 45 s). One-off = client rebooted or network blip.                                                             |
| [!] Connection terminated for peer IP. Reconnecting...                                                       | WATCH    | Client-side equivalent of the above.                                                                                                                         |
| [-] KEX Failed for IP: <err>. Retrying in Ns...                                                              | WATCH    | Client cannot complete key exchange; retries with backoff (5 s doubling to max 300 s).                                                                       |
| [*] Replacing stale session for net_id N                                                                     | INFO     | Client reconnected from a new address/port; server cleaned up the old session. Normal after client restarts.                                                 |
| [-] Drop {}: Untrusted peer authorization handshake rejection.                                               | ACT      | A peer completed the handshake but its identity is not in the approved-keys directory. Either an enrollment error (§3.3) or an unknown party - verify which. |

| Message                                                                   | Severity | Meaning / Action                                                                                                                                                                                        |
|---------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>[-] Device Starvation: Interface allocation limits exceeded.</code> | ACT      | All 255 qnx* interface slots are taken. Stale interfaces from a crash: <code>ip link delete dev qnxN</code> . Escalate if it recurs.                                                                    |
| <code>net_id pool exhausted (254 in use)</code>                           | ACT      | Server is at its 254-client limit.                                                                                                                                                                      |
| <code>[-] Tunnel interface config rejected: &lt;err&gt;</code>            | ACT      | <code>ip addr / ip link</code> failed - usually an address conflict with an existing interface or a half-torn-down qnx* device. Check <code>ip addr</code> for stale <code>10.x.y.z</code> assignments. |
| <code>[!] RX reorder overflow: skipping tickets A..B</code>               | WATCH    | Extreme packet reordering; some packets dropped. Occasional under severe network stress; frequent → underlay network problem.                                                                           |
| <code>[-] Worker dispatch error: &lt;err&gt;</code>                       | ACT      | Data-plane setup failed (usually a UDP port bind conflict). Check nothing else occupies the data port.                                                                                                  |
| <code>[LOG_ERR] ... on stdout</code>                                      | ACT      | <code>/var/log/qnyx.log</code> is not writable. Fix permissions/disk.                                                                                                                                   |

## 7. Maintenance

### 7.1 Adding / removing clients

Live operation, no restart: drop .pub files into (or remove from) the server's approved directory.

### 7.2 Capacity limits

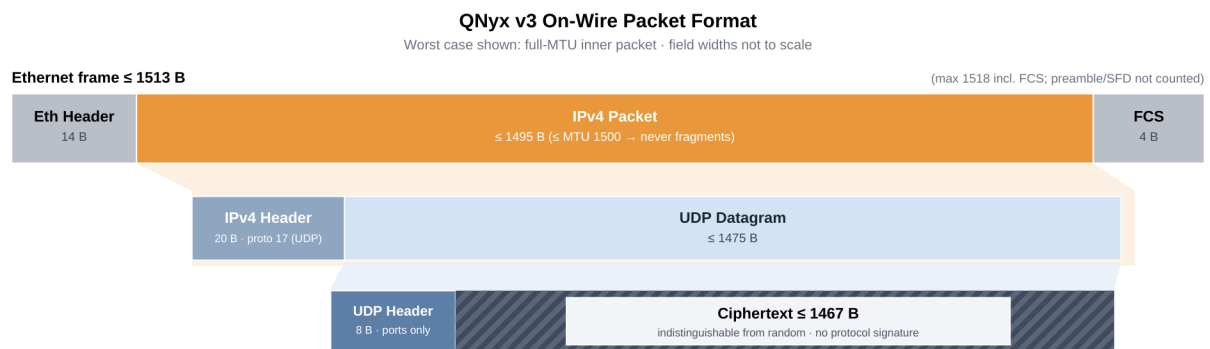
- Max 254 clients per server instance; max 255 qnx\* interfaces per box.

### 7.3 Tunnel MTU

- Tunnel MTU is defaulted at 1410 bytes, use -mtu to configure between 600-1410. Protocol and encryption add fixed 57 bytes overhead. (1410+57=1467)

### 7.4 Packet Structure

- QNyx employs a high entropy wire image with zero plaintext protocol headers.



QNyx 3.0.2 Data Plane Wire Image



## 8. Quick Reference Card

| Parameter                         | Value                                                             |
|-----------------------------------|-------------------------------------------------------------------|
| KEX channel                       | TCP, default 443                                                  |
| Data channel                      | UDP, client-chosen or -dp                                         |
| Tunnel MTU                        | 1410, or chosen with -mtu (600-1410)                              |
| Fixed overhead                    | 57 bytes → max on-wire encoded packet 1467 bytes at default MTU   |
| Heartbeat interval / timeout      | 15 s / 45 s                                                       |
| Rekey interval                    | 15 min (missed rekey → next cycle; 30-min epoch is acceptable)    |
| Old-key grace after rekey         | 60 s                                                              |
| Client KEX retry backoff          | 5 s → ×2 → cap 300 s                                              |
| Client reconnect delay after drop | 2 s                                                               |
| Tunnel addressing                 | 10.<server_id>.<net_id>.1 (server) ⇔ .2 (client), /30             |
| Limits                            | 254 clients per server · 255 interfaces per box, server IDs 0–255 |
| Log                               | /var/log/qnyx.log                                                 |
| Client routing                    | Use -r flag to route client traffic through the tunnel.           |